IsiZulu AI-Detection Classifier — Dataset Quality Report

Five quality checks run on train, eval, and test splits | 26 May 2026

Before training a classifier, it is important to ask: *is this data trustworthy?* If the training data has hidden copies of test examples, or if human and machine texts look nearly identical, the model's reported accuracy could be misleading. This report documents five checks run on the dataset and what was found.

Class Balance ~50/50 human vs machine across all splits	**Text Length** Consistent length distribution across splits	**Data Leakage** 35 exact training texts appear in eval/test (human only)	**Near-Duplicates** Small % of eval/test texts are very similar to training	**Class Separation** Human and machine texts are clearly distinct from each other

1. Class Balance ✓ Pass

A good classifier needs roughly equal examples of each class. If one class dominates, the model learns to predict it by default rather than learning genuine differences.

Train (16 377)

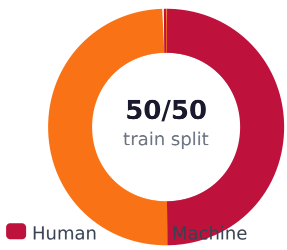
■ Human: 8 207 (50.1%) ■ Machine: 8 170 (49.9%)

Eval (1 991)

■ Human: 994 (49.9%) ■ Machine: 997 (50.1%)

Test (2 022)

■ Human: 1 013 (50.1%) ■ Machine: 1 009 (49.9%)

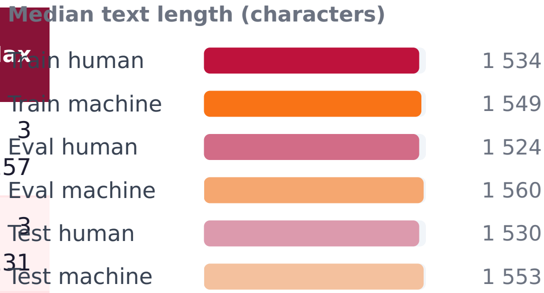


All three splits are near-perfectly balanced at 50/50. Accuracy and F1 scores are directly comparable and not inflated by class skew.

2. Text Length Distribution ✓ Pass

If texts in training are much longer or shorter than texts in the test set, the model might learn length as a proxy signal rather than genuine writing style.

Split / Class	Mean chars	Median	Std dev	Min	Max
Train — Human	1 492	1 534	206	206	3 157
Train — Machine	1 432	1 549	310	120	3 131
Eval — Human	1 479	1 524	193	314	2 164
Eval — Machine	1 438	1 560	315	208	1 865
Test — Human	1 490	1 530	201	314	1 982
Test — Machine	1 443	1 553	302	285	2 543



Bars are nearly identical across splits — no length bias.

Statistical shift test (Mann-Whitney U): No significant length shift between splits at the dataset level ($p = 0.84$ train↔eval, $p = 0.55$ train↔test). Machine texts have more variable lengths (std ~310) than human texts (std ~200) — machine generation produces more length diversity.

3. Data Leakage ⚠ Minor

Data leakage means the model saw the answer during training. If an exact copy of a test text was in the training set, the model may have memorised it rather than understood it — making the reported test score slightly over-optimistic.

Exact duplicates within each split

Split	Exact internal duplicates	% of split
Train (16 377)	**95**	**0.58%**
Eval (1 991)	**32**	**1.61%**
Test (2 022)	**7**	**0.35%**

Exact texts appearing in both train and another split

Pair	Class	Leaked	% of smaller split
Train ↔ Eval	Human	**17**	**1.71%**
Train ↔ Eval	Machine	**0**	**0.0%**
Train ↔ Test	Human	**18**	**1.78%**
Train ↔ Test	Machine	**0**	**0.0%**
Eval ↔ Test	Both	**0**	**0.0%**

What this means: 17–18 human-written texts from the training set appear word-for-word in the evaluation and test sets. This is a minor data handling issue, likely caused by how the dataset was split. The leaked texts are all human-written — there is zero leakage of machine-generated texts.

Impact on results: At 1.7–1.8% of the affected split, the effect on reported metrics is small. For a classifier achieving 99.3% accuracy, even if all 18 leaked texts benefited from memorisation, the accuracy impact is less than 0.1 percentage points. Results remain valid but should be noted.

Positive finding: Machine-generated texts show zero exact leakage across all split combinations. The model's ability to detect AI-written text is not inflated by memorisation.

4. Near-Duplicate Cross-Split Contamination ⚠ Minor

Texts that are not identical but are very similar (e.g. same article with minor edits) can still give the model an unfair advantage. Similarity is measured using a *MinHash Jaccard* score on 5-word sequences — a score of 1.0 = identical, 0.9 = almost identical, 0.5 = loosely similar.

% of eval / test texts with a near-match in training

13.4% Train↔Eval ≥ 0.5 similar	**19.9%** Train↔Test ≥ 0.5 similar	**1.1%** Eval↔Test ≥ 0.5 similar
4.1% Train↔Eval ≥ 0.7 similar	**6.6%** Train↔Test ≥ 0.7 similar	**0.3%** Eval↔Test ≥ 0.7 similar
1.5% Train↔Eval ≥ 0.9 similar	**3.8%** Train↔Test ≥ 0.9 similar	**0.0%** Eval↔Test ≥ 0.9 similar

Deeper orange = higher contamination risk.

How to read this: At the loose threshold (≥ 0.5), up to 20% of test texts share some similarity with training. But "loosely similar" for IsiZulu texts of ~1 500 characters is expected — the language has a limited vocabulary and texts share common phrases. At the strict threshold (≥ 0.9 = almost identical), only 3.8% of test texts are affected.

Key reassurance: Eval and test are almost completely clean with respect to each other (0–1.1%). This means the two are truly independent. The near-duplicate overlap is primarily a train↔test concern at loose thresholds, not a cross-evaluation contamination concern.

5. Class Separation — Are Human and Machine Texts Distinct? ✓ Pass

If a human text and a machine text are nearly identical, labelling becomes ambiguous and the classifier cannot reliably distinguish them. This check looks for any human text that closely resembles a machine text *within the same split*.

Similarity threshold	Cross-class pairs (train)	Cross-class pairs (eval)	Cross-class pairs (test)
≥ 0.5 (loosely similar)	1	0	0
≥ 0.6	0	0	0
≥ 0.7	0	0	0
≥ 0.8	0	0	0
≥ 0.9 (almost identical)	0	0	0

Only 1 cross-class pair was found anywhere in the entire dataset (at the loosest threshold). Human texts and machine texts are overwhelmingly distinct from each other. The labels are unambiguous.

Split-to-split domain similarity (cosine, 0–1)

Pair	Similarity	Meaning
Train ↔ Eval	**0.9913**	Same domain ✓
Train ↔ Test	**0.9923**	Same domain ✓
Eval ↔ Test	**0.9842**	Same domain ✓

Domain consistency: All three splits are drawn from the same underlying data distribution. The model is not tested on a domain it has never seen — a common cause of overly-optimistic results in NLP research.

Overall Dataset Quality Verdict

Check	Finding	Risk to results	Status
Class balance	50/50 across all splits	None — metrics are unbiased	✓ **Pass**
Text length consistency	Mean ~1 460 chars, no significant shift	None — no length shortcut available	✓ **Pass**
Exact data leakage	35 human texts leaked train→eval/test (<1.8%)	Low — <0.1 pp impact on accuracy	⚠ **Minor**
Near-duplicate contamination	1.5–3.8% of test texts very similar to train (≥0.9)	Low to moderate — test score slightly optimistic	⚠ **Minor**
Cross-class separation	1 ambiguous pair in 16 377 training examples	Negligible — labels are clean	✓ **Pass**
Domain consistency	All splits from same distribution (similarity 0.98–0.99)	None — fair evaluation conditions	✓ **Pass**

Summary: The dataset is of good overall quality. The class balance is excellent, text lengths are consistent, and human and machine texts are clearly distinct — all conditions for reliable training and evaluation. Two minor concerns were identified: a small number of exact human texts leaked from training into evaluation and test sets (<1.8%), and a modest fraction of test texts have close near-duplicates in training at loose similarity thresholds. These do not invalidate the results, but mean the reported accuracy may be marginally over-optimistic by roughly 0.1–0.5 percentage points.

Quality checks: general_leakage_test/, cross_class_similarity/, dataset_leakage_test/ | Method: MinHash Jaccard (5-grams, 128 permutations), Mann-Whitney U test, cosine similarity | 26 May 2026